Hong–Ou–Mandel Interferometer

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1 Introducton

This document describes a modular baseplate and a set of optomechanical mounts designed for the Hong–Ou–Mandel (HOM) interference. The central design goal was that every optical axis in the interferometer land on a standard hole of the optical bench, so that individual mounts can be added, removed, or repositioned without re-referencing the rest of the layout. Each part was first prototyped by 3D printing to verify fit, then finalized as a CNC-machined component through an external vendor. Section 5- 7 describes about the commercially available optics that the mount holds, geometry about the hole patterns relative to the beam, and key dimensions necessary to reproduce or adapt it. A reader who only wants to utilize or modify one mount can jump directly to its section.

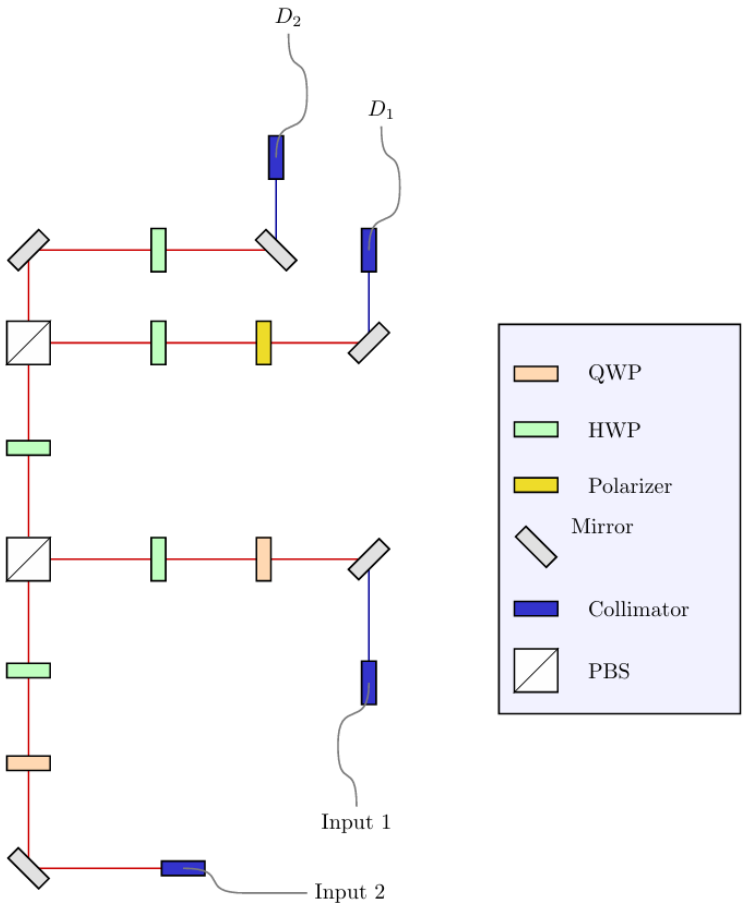


Figure 1: Proposed Design

2 Background: Hong–Ou–Mandel Interference

When two indistinguishable heralded single photons enter the two input ports of a 50:50 beamsplitter, they leave together through the same output port. In such scenarios, the coincidence of finding photons in different output ports drops to zero, which is represented as the characteristic HOM "dip". This interferometer is designed to use an external delay line to control the time delay and make the photons perfectly indistinguishable in time; tuning the delay traces out the HOM dip curve that this setup is built to observe. The subsections below are not a full derivation of the HOM effect; they are an equational walkthrough of how this particular device prepares the two input photons and what, in that signal chain, produces the HOM dip. For treatments of HOM interference in the literature—including how spectral and temporal distinguishability enter the dip—see Brańczyk [4] and Drago and Brańczyk [5].

2.1 Polarization state from the fiber

Each heralded single photon arrives input ports through a single-mode fiber, generally leaving it in an elliptical polarization state. In the horizontal/vertical basis,

$$\left|\psi_{\text{fib},a}\right\rangle = \cos\gamma\,|H\rangle + e^{i\delta}\sin\gamma\,|V\rangle = \begin{pmatrix}\cos\gamma\\ e^{i\delta}\sin\gamma\end{pmatrix}. \tag{1}$$

Here, δ is the relative phase responsible for elliptical polarization.

2.2 Polarization correction: QWP + HWP

A quarter-wave plate (QWP) followed by a half-wave plate (HWP) at certain angles will help resolve the elliptical polarization state to a known linear axis. The unitary for QWP is given as:

$$U_{\text{QWP}}(\phi) = \begin{pmatrix}\cos^2\phi + i\sin^2\phi & (1-i)\sin\phi\cos\phi\\ (1-i)\sin\phi\cos\phi & \sin^2\phi + i\cos^2\phi\end{pmatrix}. \tag{2}$$

When the unitary acts on the state $\left|\psi_{\text{fib},a}\right\rangle$ at an angle ϕ_c,

$$U_{\text{QWP}}(\phi_c)\left|\psi_{\text{fib},a}\right\rangle = \begin{pmatrix}\cos\gamma\\ \sin\gamma\end{pmatrix}. \tag{3}$$

The unitary for HWP is given by:

$$U_{\text{HWP}}(\theta) = \begin{pmatrix}\cos 2\theta & \sin 2\theta\\ \sin 2\theta & -\cos 2\theta\end{pmatrix}. \tag{4}$$

Choosing θ so that 2θ rotates γ onto the vertical axis sends the elliptical input mode to a clean vertical mode:

$$a^\dagger_{\text{elliptical}} \xrightarrow{\text{QWP+HWP}} a^\dagger_V. \tag{5}$$

Similarly choosing β for horizontal mode:

$$b^\dagger_{\text{elliptical}} \xrightarrow{\text{QWP+HWP}} b^\dagger_H. \tag{6}$$

2.3 Combining the arms at the beamsplitter

With both photons prepared, the combining stage acts as the interfering beamsplitter.

$$U_{\text{PBS}} = \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}, \tag{7}$$

and a HWP set to 22.5° acts as the (50:50) beam splitter resulting in interference.

$$U_{\text{HWP}}(22.5^\circ) = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}. \tag{8}$$

In creation-operator form, the two input modes ($b_H^\dagger$ and $a_V^\dagger$) map to the two output modes ($c_H^\dagger$ and $d_V^\dagger$) as

$$\begin{pmatrix} c_H^\dagger \\ d_V^\dagger \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} b_H^\dagger \\ a_V^\dagger \end{pmatrix}, \tag{9}$$

so that

$$b_H^\dagger \to \frac{1}{\sqrt{2}}\left(c_H^\dagger - d_V^\dagger\right), \qquad a_V^\dagger \to \frac{1}{\sqrt{2}}\left(c_H^\dagger + d_V^\dagger\right). \tag{10}$$

The mixing of the two input modes then produces the two-photon interference which will be measured as the HOM dip at the two detectors.

2.4 Interference and coincidence cancellation

The prepared input state is

$$|\psi_{\text{in}}\rangle = a_V^\dagger b_H^\dagger |\text{vac}\rangle . \tag{11}$$

Substituting the mode mapping,

$$|\psi_{\text{out}}\rangle = \frac{1}{\sqrt{2}}\left(c_H^\dagger + d_V^\dagger\right) \frac{1}{\sqrt{2}}\left(c_H^\dagger - d_V^\dagger\right) |\text{vac}\rangle . \tag{12}$$

Expanding,

$$|\psi_{\text{out}}\rangle = \frac{1}{2}\left(c_H^\dagger c_H^\dagger - c_H^\dagger d_V^\dagger + d_V^\dagger c_H^\dagger - d_V^\dagger d_V^\dagger\right) |\text{vac}\rangle . \tag{13}$$

At $\tau = 0$ the output operators commute, $[c_H^\dagger, d_V^\dagger] = 0$, so the cross terms cancel:

$$-c_H^\dagger d_V^\dagger + d_V^\dagger c_H^\dagger = 0. \tag{14}$$

Hence

$$|\psi_{\text{out}}\rangle = \frac{1}{2}\left((c_H^\dagger)^2 - (d_V^\dagger)^2\right) |\text{vac}\rangle = \frac{1}{\sqrt{2}} |2_c, 0_d\rangle - \frac{1}{\sqrt{2}} |0_c, 2_d\rangle . \tag{15}$$

Projecting onto the coincidence state $|1_c, 1_d\rangle$,

$$P_c = \left|\langle 1_c, 1_d \,|\, \psi_{\text{out}}\rangle\right|^2 = 0. \tag{16}$$

The coincidence rate vanishes: the HOM dip reaches zero.

2.5 Motivation: the reference setup and a compact HOM device

The motivation for this device set is the HOM setup of Michael Grayson and team [1], the design the lab currently uses and the starting point this project builds on (Fig. 2); it connects to the group's wider work on high-quality entangled-photon sources [3] and phase-stable quantum-network links [2].

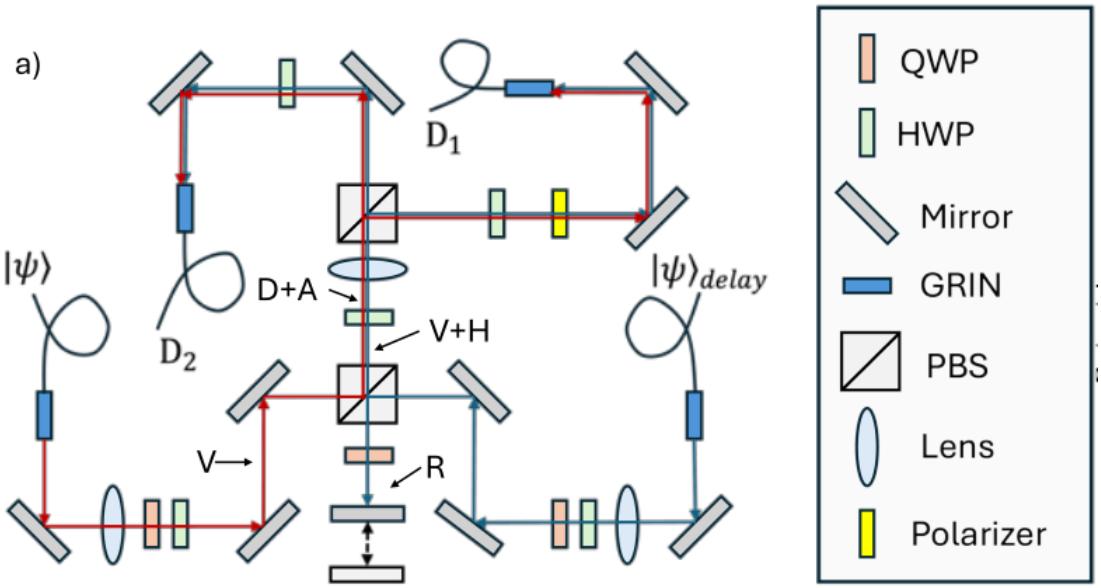


Figure 2: Michael's Setup

This project aims to make Michael's setup more compact by moving the interferometer from individually positioned mounts on the optical bench to a common modular baseplate. The major changes in Figure 1, compared to Figure 2, are the removal of the intermediate lenses, a shorter total optical path, and the separation of the delay stage into a standalone project. Removing the intermediate lenses reduces the number of optical components that need to be mounted and aligned, while the shorter beam path allows the remaining optics to fit within a smaller footprint.

The delay stage remains necessary to control the relative arrival time of the two photons, but it is treated as an external module whose design can be optimized separately to reduce optical loss. This allows the present project to focus on the compact interferometer and its mounts, while leaving the development of a lower-loss delay stage to a separate effort.

3 System Layout and Beam Path

The total optical path from either input to a detector is 18 inches, with 8 inches from either input to the first PBS, 3 inches in between the PBSs, and 7 inches from the second PBS to either detector. The bottom two starting points are the incoming beams, and the top two end points are the outgoing beams.

The interferometer is designed to be compact, with a nominal distance of 6 inches from each input to the first PBS. These nominal input locations are labeled **Input 1** and **Input 2** in Figure 3. However, the two inputs are collimated along different axes—one horizontal and one vertical. At the nominal input locations, the two collimator setups can therefore intersect, which can make collimation difficult.

To avoid this crossing, the right input is rerouted by an additional 1 inch vertically and 1 inch horizontally. This turns its coupling line so that it runs horizontally alongside the other input rather than crossing it. The rerouting adds a total of 2 inches to the optical path, which is accommodated by extending the bottom of the layout by 2 inches. The corresponding extended input locations are labeled **Input 1′** and **Input 2′**. Thus, the nominal distance remains 6 inches at **Input 1** and **Input 2**, while the layout can be extended to 8 inches at **Input 1′** and **Input 2′** when the coupling configuration requires the rerouted path.

Link To CAD File

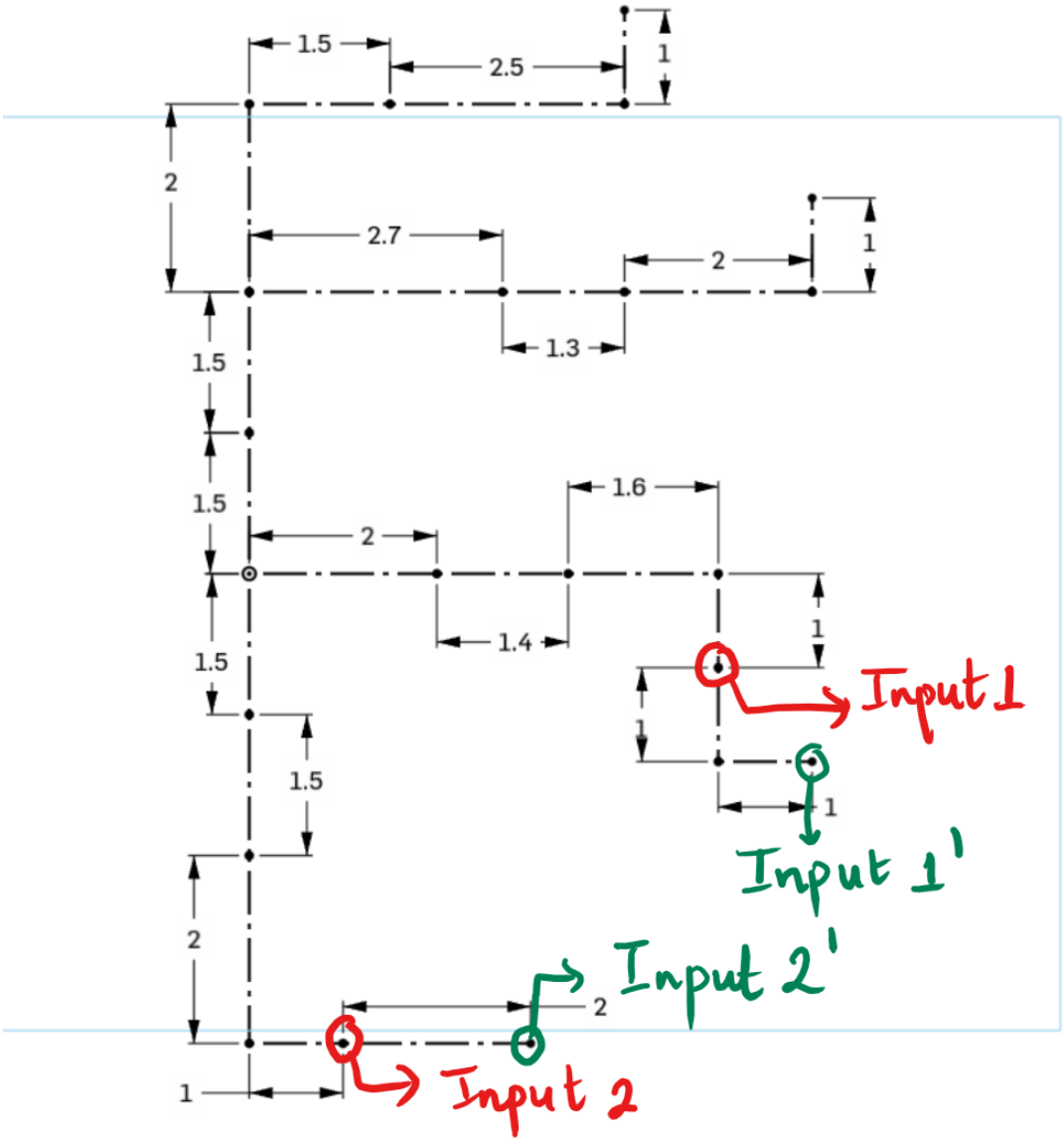


Figure 3: Base Sketch

This is the 2D reference for the whole interferometer; every mount is ultimately tied back to a point on it. The circled dot (⊙) near the left is the sketch origin. Each remaining solid dot is the optical-axis center of one optic — the point where the beam meets that optic — and the dash-dot lines are the beam axes joining them. Read the drawing as a folded beam path: a vertical spine runs up the left side, and at each optic a horizontal branch carries the beam across the plate. The dimensions along the spine (2, 1.5, 1.5, 1.5, 2) fix the vertical positions of the rows, and the dimensions on each branch (e.g. 1.5/2.5 on the top row; 2.7/1.3/2 on the next; 2/1.4/1.6 through the origin row) fix where each optic sits along that row. The short 1-inch stubs at the far ends of the top and upper rows are the last segments out to the input/output fiber couplers. Note that the beam-to-beam distances on *perpendicular* paths resolve to whole inches, per the spacing rule in Section 4; the fractional numbers position the reference geometry that builds those whole-inch legs.

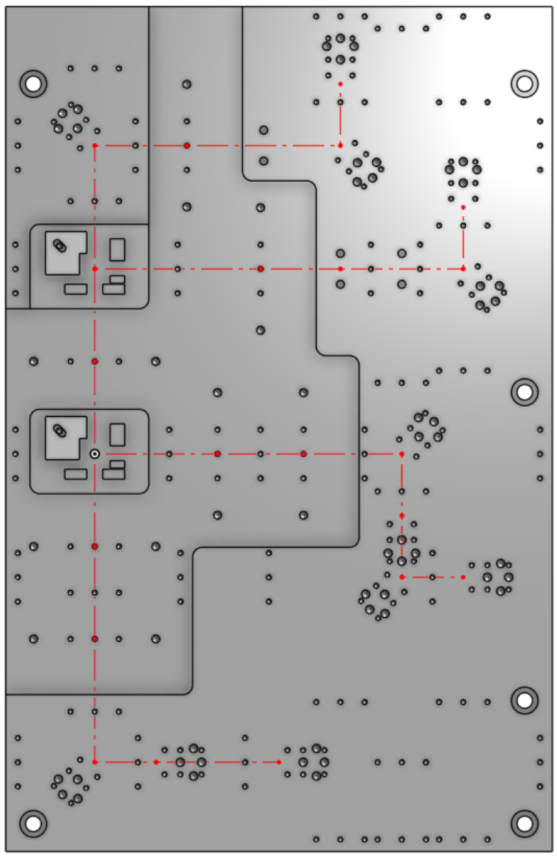


Figure 4: Top View

This is the top view baseplate seen from directly above as rendered in the CAD software, with the beam path from the base sketch overlaid as red dash-dot lines so you can see how the abstract sketch maps onto real hardware. The dense field of small holes are the collections of tapped and through hole grid that lets every optic land on a bench hole (the whole-inch rule in practice). The two rounded-rectangle recesses on the left half are the seats for the two raised blocks that carry the polarizing beam splitters explained in Section 6. The four large counterbored holes near the corners are the through-holes that bolt the whole plate down to the optical table.

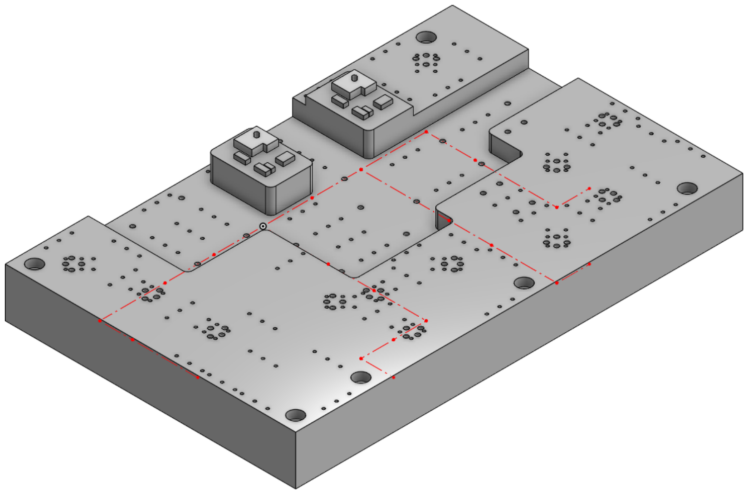


Figure 5: Isometric View

The same plate rendered in isometric view, which makes the *beam heights* visible.

4 Design Conventions

These conventions are the reusable core of the design. Anyone adding or swapping a mount should follow them so the layout stays self-consistent and every optic keeps landing on the bench grid.

4.1 Point of incidence versus the vendor's reference edge

Every dot in the base sketch (Figure 3) marks a *point of optical incidence* — the exact location where the incoming beam meets the optic surface. For a mirror this is the center of the coated surface. These points, not the mounts, define the whole-inch grid, because it is the beam path that must stay aligned to the bench. **The point of incidence is the fixed anchor and never moves.**

Commercial mounts, however, are not always dimensioned from the point of incidence. Mostly, holes are referenced from a single physical face of the mount body — one fixed edge. Placing such a mount in the design therefore takes two steps: *the reference edge is offset from the fixed point of incidence by the optic's known geometry, and from that edge every hole and feature is laid out exactly as the vendor specifies.* In short:

$$\text{point of incidence (fixed)} \longrightarrow \text{offset to the mount edge} \longrightarrow \text{holes referenced from that edge.}$$

The offset is what bridges the two reference frames — the beam's frame (point of incidence) and the vendor's frame (mount edge). Because a single edge is the zero for most of the vendor's dimensions, the point of incidence simply sits at a fixed distance from that zero; nothing about the incidence point changes when the mount is placed.

4.2 Whole-inch spacing rule

The distance between any two optical-axis centers that lie on perpendicular beam paths is always a whole number of inches. This keeps every optical axis aligned to the tapped-hole grid of the optical bench, so additional optics can be inserted anywhere along a beam path — through the bench holes — without disturbing the existing layout.

4.3 Front- versus rear-loading mounts

Whether an optic loads from the front or the rear of its mount determines where the optical surface sits relative to the mount body, and therefore the size of the offset in Section 4. A **rear-loading** mount (Opto-Sigma, Section 5.1)

places the optical surface a fixed distance behind the front face, independent of the optic, so the offset is a constant. A **front-loading** mount (Polaris, Section 5.2) references the optic from the front, so the surface position — and therefore the offset — changes with the mirror thickness. This is exactly why the two Polaris variants use different references (+4.4 mm for a 3 mm mirror, −4.6 mm for a 12 mm mirror): the incidence point is fixed, and the thicker mirror pushes the reference edge to a different offset.

5 Mirror Mounts

5.1 Opto-Sigma Half-Inch Mirror Mount (VMHV-12.7L-R)

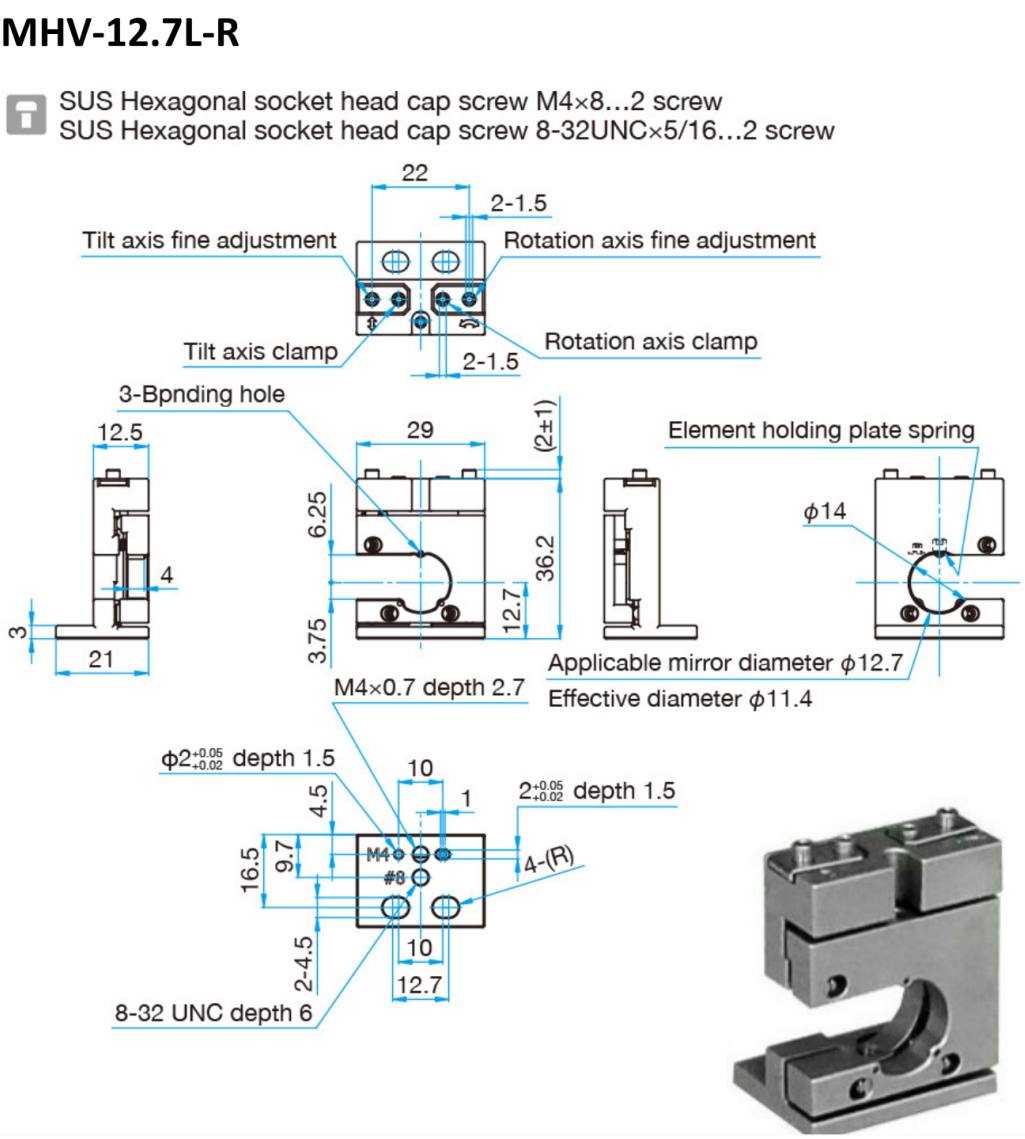


Figure 6: Opto-Sigma VMHV-12.7L-R

This is Opto-Sigma's CAD drawing, and the main dimensions we need from it are the following. The mount accepts a ∅12.7 mm (half-inch) mirror with a ∅11.4 mm effective (clear) aperture. The mounting-hole pattern on the base uses two 8-32 UNC holes on a 12.7 mm square, along with two ∅2 mm alignment-pin holes, and the tilt/rotation adjusters are located on the top face.

This is a rear-loading mount, so the mirror surface sits at a fixed distance behind the front face regardless of the optic. As shown in the CAD file, the front face lies 4.5 mm from the pin holes and 16.5 mm from the screw holes. The mirror sits 1 mm behind the front face held by the feature labeled **3-Bending Hole**. However, the thickness of the feature was not provided on the drawing file, and was measured using the vendor's official 3D step file. Applying the referencing rule of Section 4: the point of incidence is fixed, the reference edge (front face) is offset 1 mm from it, and the vendor's holes are then measured from that edge. The mirror surface is therefore $4.5\ \text{mm} - 1\ \text{mm} = 3.5\ \text{mm}$ from the pin holes and $16.5\ \text{mm} - 1\ \text{mm} = 15.5\ \text{mm}$ from the screw holes. The sketch is referenced to the base-sketch incident point using these two offsets.

Link To The Product

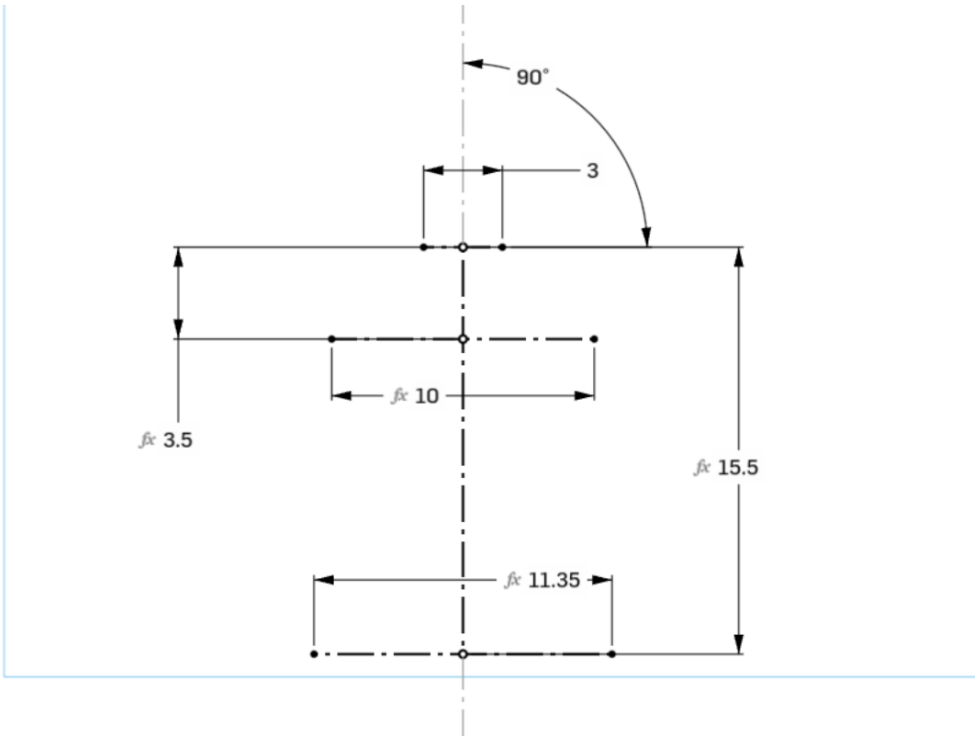


Figure 7: Opto-Sigma Mount Sketch

This is the sketch that places the mount's holes relative to the beam. The vertical dash-dot line is the beam axis, and the '3" dimension defines a short construction line used only to build the pattern — it is not a hole spacing, and the circular dot on it is the point of incidence (the origin, tied to the base sketch). Working down from the top: the 90° callout and the small '3.5' (left) indicate the vertical distance from the point of incidence down to the pin-hole row, and 15.5 (right) is the distance down to the screw-hole row — the two offsets derived below. 10 is the horizontal spacing of the pin-hole pair, and 11.35 is the horizontal spacing of the screw-hole pair. Each solid dot is a hole center.

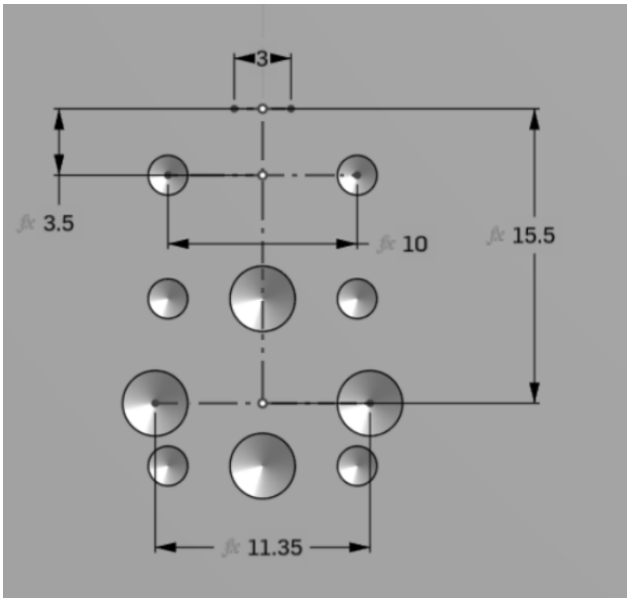


Figure 8: Opto-Sigma 3D Rendered View

This figure shows how the different holes in the sketch corresponding to the holes on the machined mount. The upper pair, spaced 10 mm apart, are the alignment-pin holes, while the lower pair of larger holes, spaced 11.35 mm apart and located 15.5 mm below the incidence point, are the tapped screw holes. The remaining holes are for other mirror mounts discussed in the following sections.

5.2 Polaris Mirror Mount (POLARIS-K05VS2)

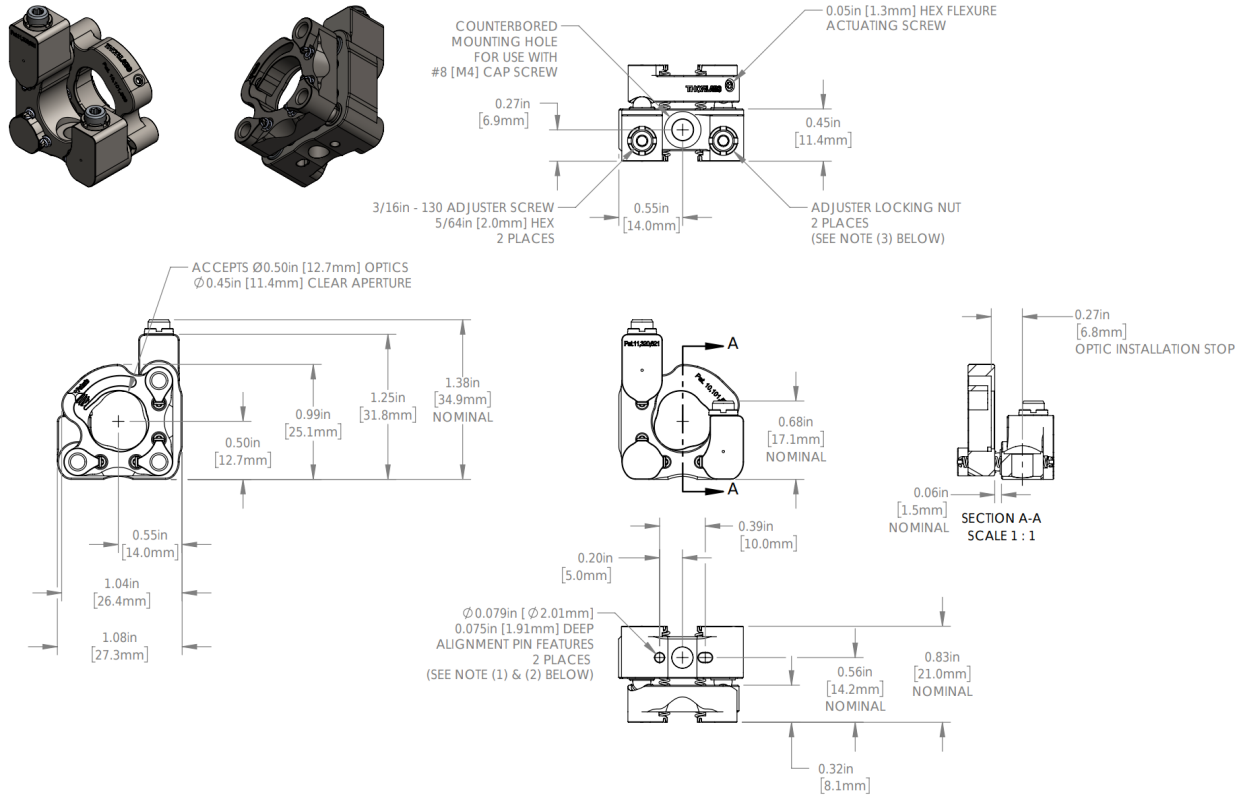


Figure 9: POLARIS-K05VS2

Link To The Product

From Thorlabs' drawing provides the following relevant information: the mount is front-loading and accepts ∅0.50 in (∅12.7 mm) optics with a ∅0.45 in clear aperture; it bolts down through a tapped hole for a #8 (M4) cap screw; and it has two alignment-pin features (∅0.079 in, 0.075 in deep).

Since the Polaris mounts are front-loading, the thickness of the mirror determines where the optical surface sits relative to the mount and therefore how the sketch references the base-sketch incident point. Drawing the sketch with the standard 6 mm thick mirrors available at the lab would result in a collision of holes between the Polaris mounts and the Opto-Sigma mounts. So, 3 mm and 12 mm thicknesses were considered to account for two additional mount spaces without collision. The distance between the part of the mount where the mirror rests and the front of the mount was 7.4 mm. This measurement was not provided in the PDF documentation by Thorlabs, so the distance was measured using the official 3D step file.

5.2.1 Polaris with 3mm Thick Mirror

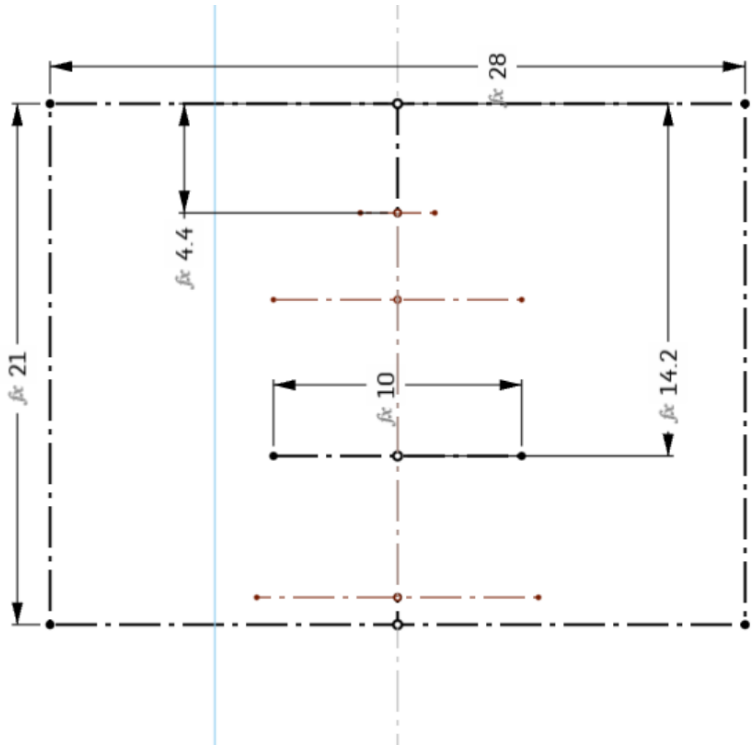


Figure 10: POLARIS sketch for 3mm thick mirror

The outer dashed rectangle is the mount-body outline (construction geometry), 28 mm wide and 21 mm tall. 4.2 mm is the offset from the top edge to the point of incidence, 14.2 mm is the vertical distance from the top edge down to the main hole row, and 10 mm is the horizontal spacing of the central hole pair. The brown dash-dot lines running through the middle are the Opto Sigma sketch lines, the same as those shown in Figure 7.

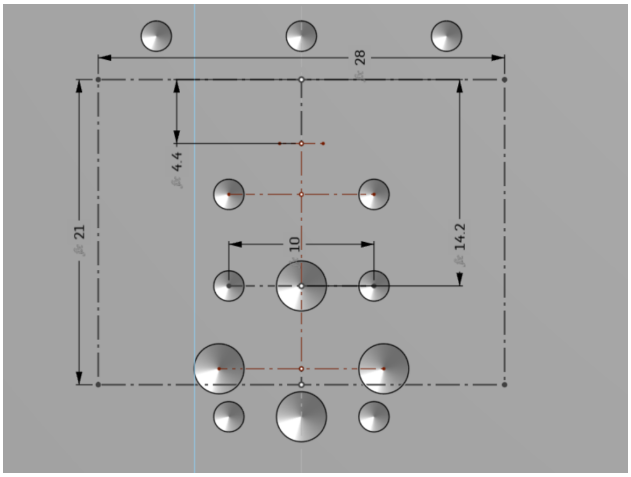


Figure 11: Polaris 3D Rendered View

The 3D-rendered version shows the same pattern: the central row with the 10 mm-spaced pair straddling a larger central clearance hole, and the lower cluster of mounting holes, all within the 28×21 mm dimension. Matching it

to the sketch, the larger circle at the middle is a tapped hole, while the smaller ones are the pin holes for alignment. When using a 3 mm-thick mirror, the surface of the mirror sits $7.4\text{ mm} - 3\text{ mm} = 4.4\text{ mm}$ inside the mount. Hence, the front of the mount (the reference edge) was placed 4.4 mm from the fixed incident beam point in the base sketch. Every other hole was then referenced to the front of the mount.

5.2.2 Polaris With 12mm Thick Mirror

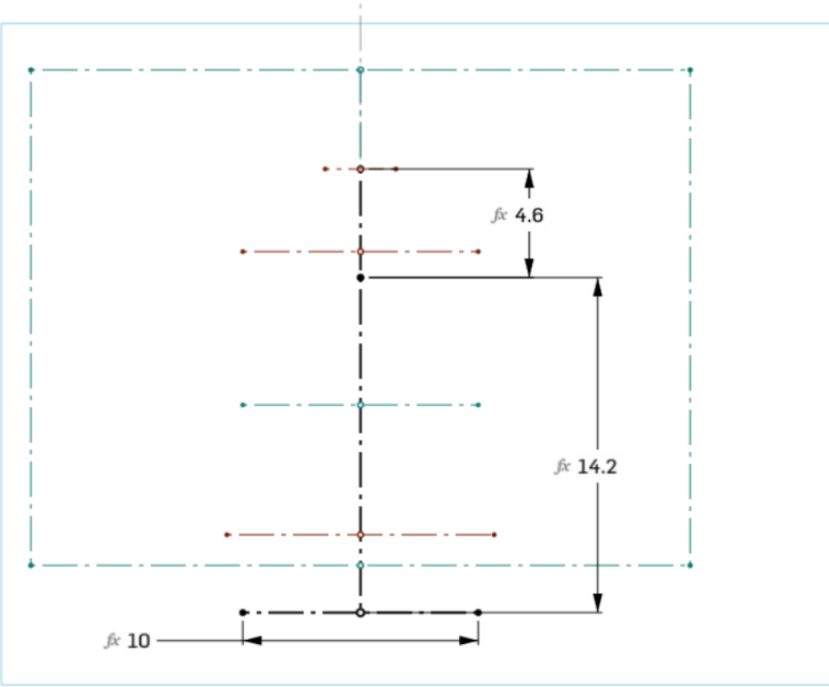


Figure 12: POLARIS with 12mm Thick Mirror Sketch

This is the same mount drawn for a 12 mm mirror, where the blue lines provides direct comparisons to the 3 mm sketch above. The 14.2 hole-row distance are unchanged, but the top offset is now 4.6 and it is applied in the opposite sense — because the thicker mirror pushes the reference edge to the *other* side of the point of incidence. That single sign flip (relative to the 3 mm case) is the whole difference between the two variants, and it is exactly why one references at $+4.4$ mm and the other at -4.6 mm.

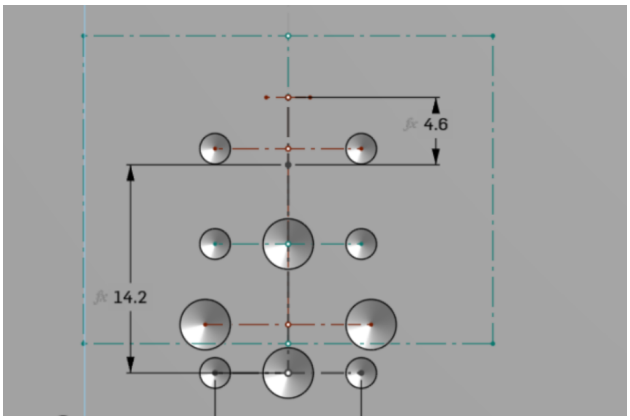


Figure 13: POLARIS 3D Rendered View

This is the 3D rendered 12 mm pattern. The hole layout is the same family as the 3 mm version shifted vertically by the different edge reference. Here, the front of the mount was placed -4.6 mm from the fixed incident beam point in the base sketch. Every other hole was then referenced to the front of the mount.

6 PBS Holder

This mount holds the polarizing beamsplitter (PBS) used to combine the two interferometer arms. The PBS is constrained between four raised locating features and a separate 3D-printed L-shaped clamp. The clamp is fastened to the mount through an M3 tapped hole and applies pressure toward the raised features, securing the PBS in place while leaving the optical paths unobstructed.

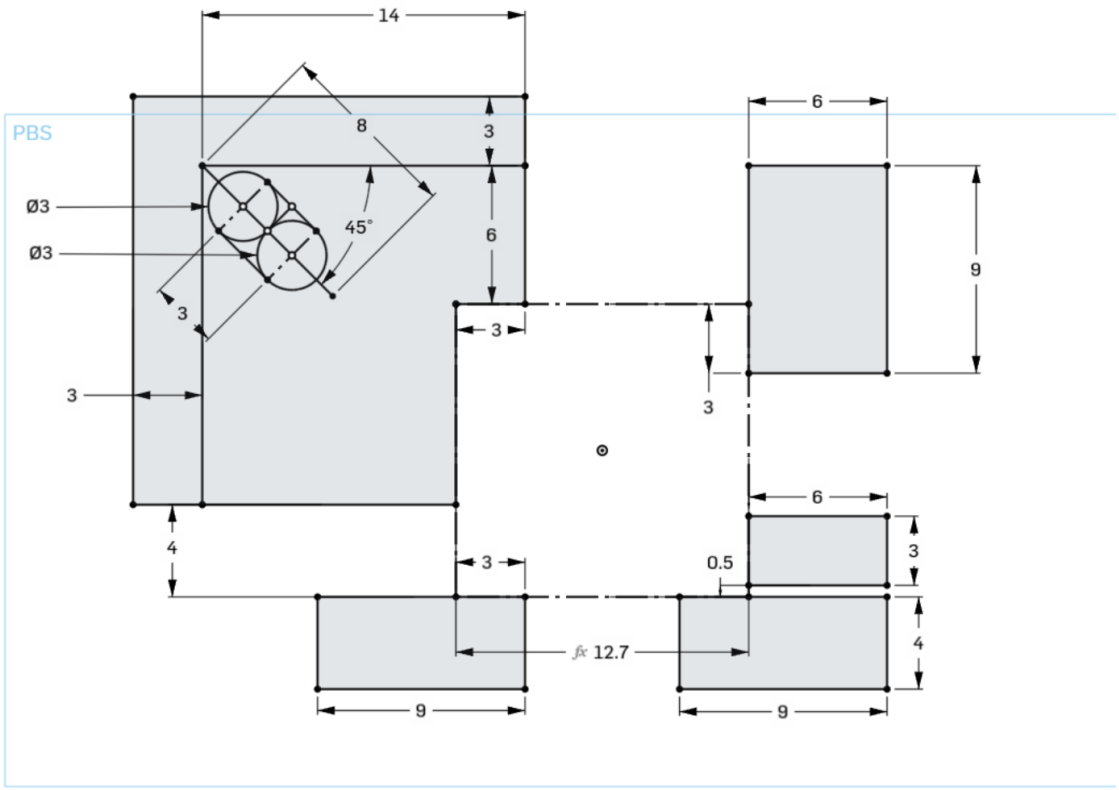


Figure 14: PBS Mount Sketch

The circled dot near the center of the sketch marks the origin projected onto the base sketch, and the PBS dimensions are based on a nominal half-inch (12.7 mm) square optic. The shaded L-shaped region in the upper-left represents the separate 3D-printed clamp used to secure the PBS against the four raised locating features of the CNC-machined mount. The two ∅3 mm circles define the ends of a capsule-shaped slot in the clamp. This slot allows the clamp to be fastened to the M3 tapped hole in the holder while providing some freedom to adjust its position during installation. The clamp is positioned diagonally relative to the PBS. As shown in the sketch, its slot is oriented at 45°, with the hole centered along the indicated 8 mm-wide reference region. The 14 mm dimension across the upper portion of the clamp was used to locate the outer edge of this feature. The additional 3 mm extension of the L-shaped clamp increases the material surrounding the clamping region and may be increased or decreased as required for a particular implementation. The four smaller shaded rectangular regions are raised features against which the PBS is held by the 3D-printed clamp. Their dimensions and positions are shown individually in the sketch: the upper-right feature is 6×9 mm, the lower-left feature is 9×4 mm, and the lower-right features are located using the indicated 6, 9, 3, 4, and 0.5 mm dimensions. A 3 mm clearance was maintained around the raised features so that the mount does not interfere with the incoming beams. These dimensions were selected iteratively by testing several 3D-printed versions of the mount and adjusting the geometry to obtain reliable constraint of the PBS while maintaining sufficient clearance for the incident and outgoing beams.

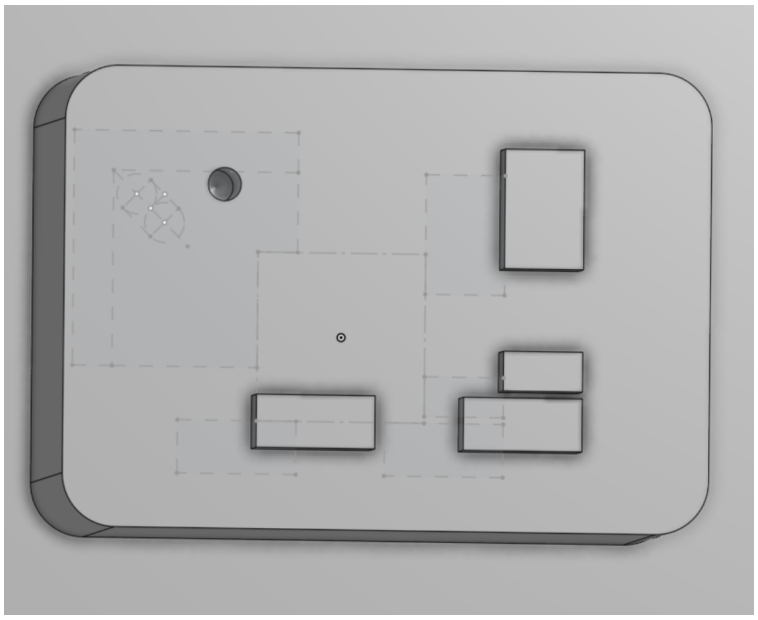


Figure 15: PBS Mount 3D Rendered View without clamp

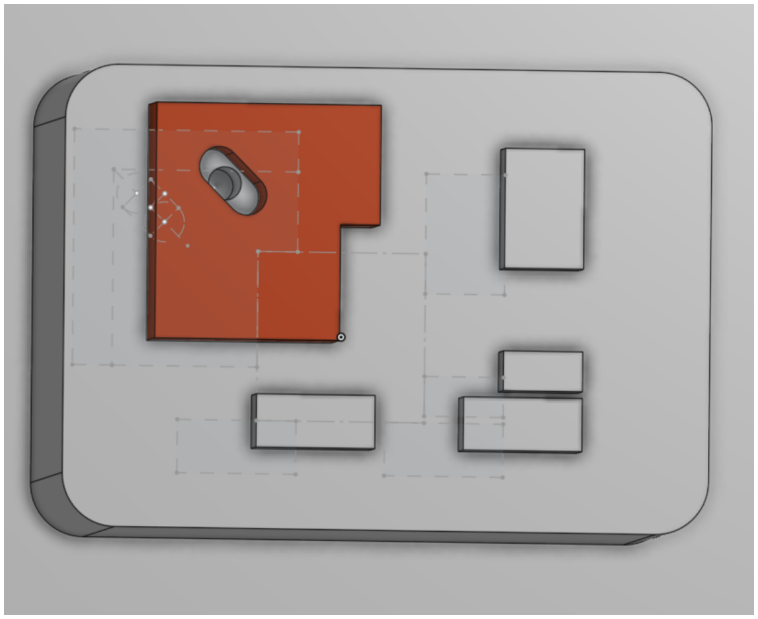


Figure 16: PBS Mount 3D Rendered View with clamp

Figures 15 and 16 show the resulting 3D rendered geometry without and with the 3D-printed clamp installed, respectively.

7 ELL16

This mount carries Thorlabs ELL16 resonant piezoelectric rotation mount, used to rotate a wave-plate under computer control for polarization setting.

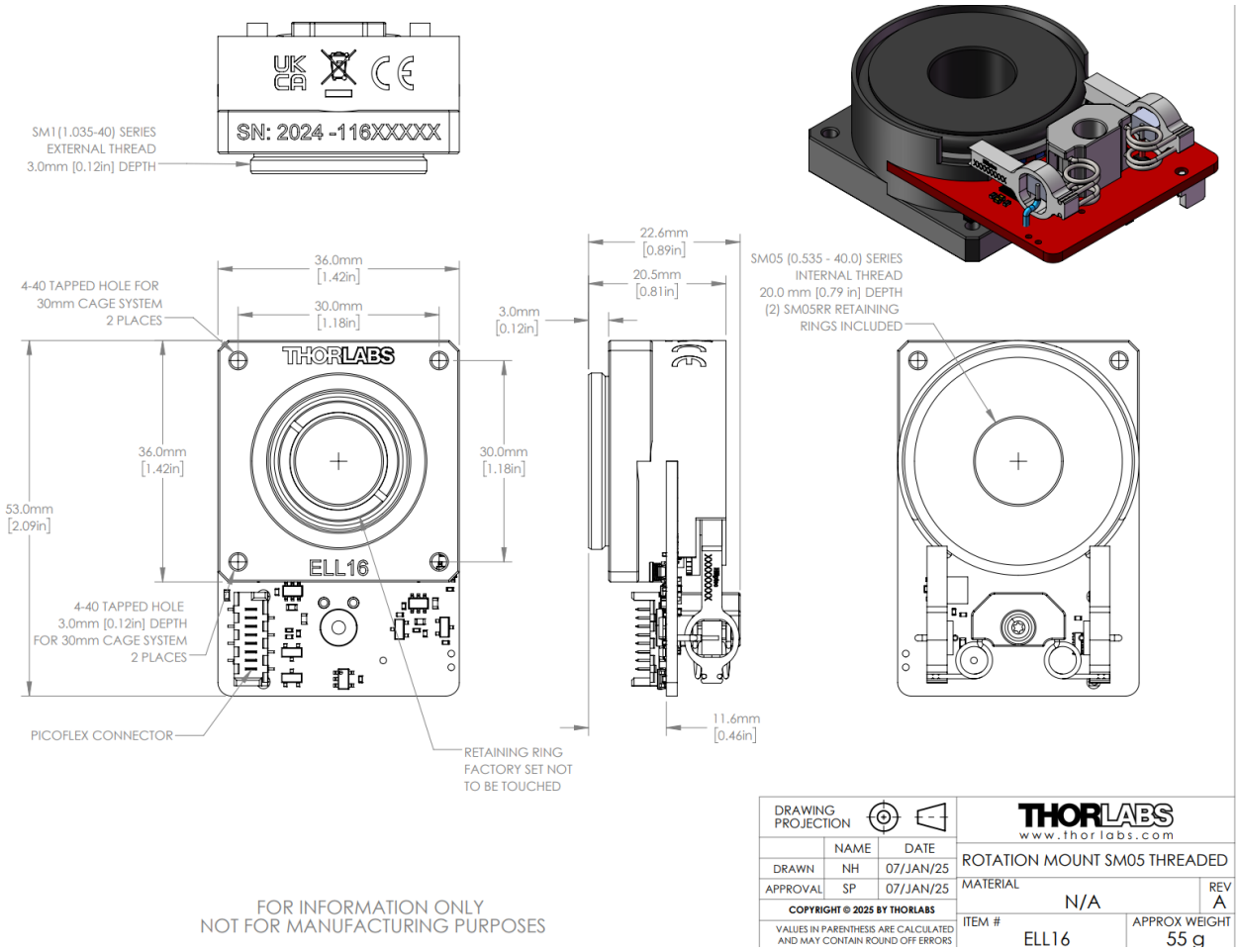


Figure 17: Ell16

Link To The Product

Thorlabs' drawing gives the interface that the custom holder must match: the ELL16 is an SM05-threaded resonant rotation mount, roughly 53×36 mm across its face, mounted through four 4-40 tapped holes arranged on a 30 mm cage-system square, with the drive electronics and Picoflex connector on one side and a factory-set retaining ring around the aperture. These four cage holes and the aperture are the primary features that the custom case must locate. Several versions of the holder were designed and tested in the lab. For the current setup, the holder must position the optical axis 1 inch above the baseplate and provide mounting holes spaced 50.5 mm apart. Optional alignment pins can also be incorporated into the design; when the rotation mount is not in use, these pins can serve as supports for an iris.

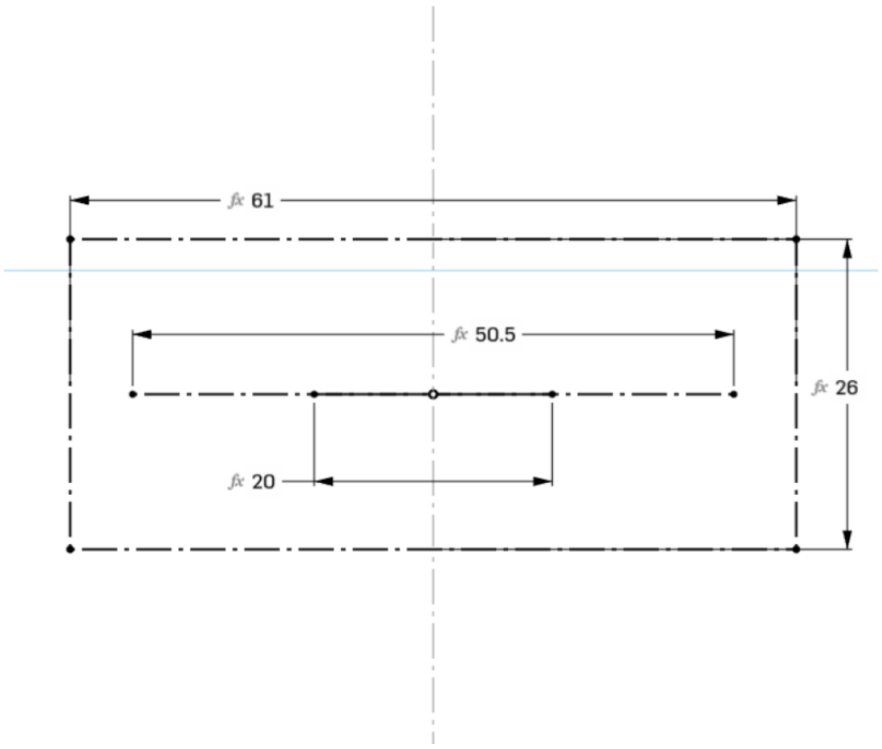


Figure 18: Ell16 Mount Sketch

The outer dashed rectangle represents the baseplate outline and also corresponds to the maximum footprint of the ELL16 holder on the mount, measuring 61 mm wide and 26 mm long. The circled dot at its center is incident to the base sketch. Five hole centers lie along the central horizontal centerline: the two outer holes are the mounting screw holes (M4-tapped), spaced 50.5 mm apart, while the three inner holes are for alignment dowel pins. The two outer dowel-pin holes are separated by 20 mm, with the center hole positioned midway between them, giving a spacing of 10 mm between adjacent dowel-pin holes.

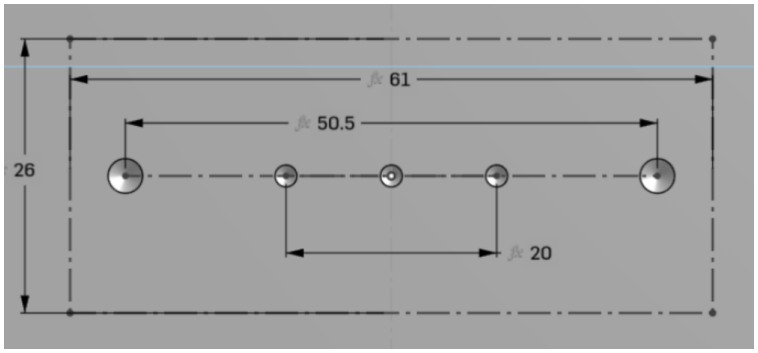


Figure 19: Ell16 Mount 3D Rendered View

The 3D rendered plate shows tapped holes for screw and drilled holes for dowel pins. As noted above, those three small holes double as iris holes when the ELL16 is removed.

7.1 Example ELL16 Holder/Case

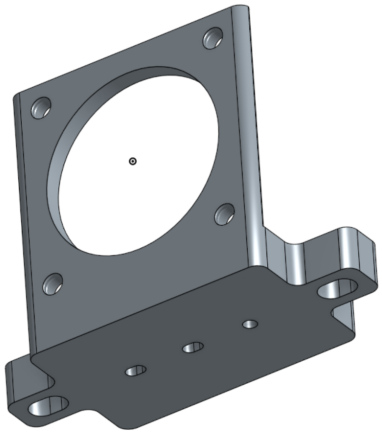


Figure 20: Ell16 Case

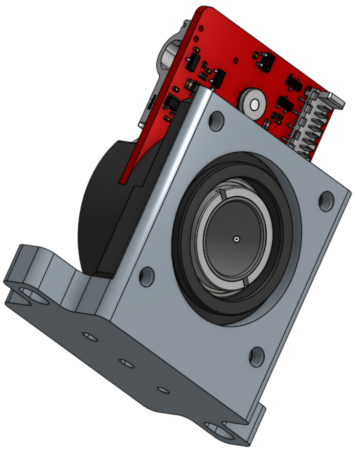


Figure 21: Ell16 Assembly

Link To The CAD File for ELL16 Holder

Figures 20 and 21 shows an example of completed ELL16 holder and the resulting assembly, respectively. The horizontal base contains the two outer mounting holes and three smaller collinear dowel-pin holes; when the ELL16 is removed, the dowel pins can also serve as supports for an iris. The two slotted mounting tabs provide a small amount of translational adjustment when fastening the holder to the baseplate. The vertical face contains the central optical aperture and four mounting holes arranged on the 30 mm cage-system square used to secure the ELL16. Figure 19 shows the ELL16 installed in the holder, with the optical aperture aligned while the drive electronics and connector remain accessible.

8 Example Assembly

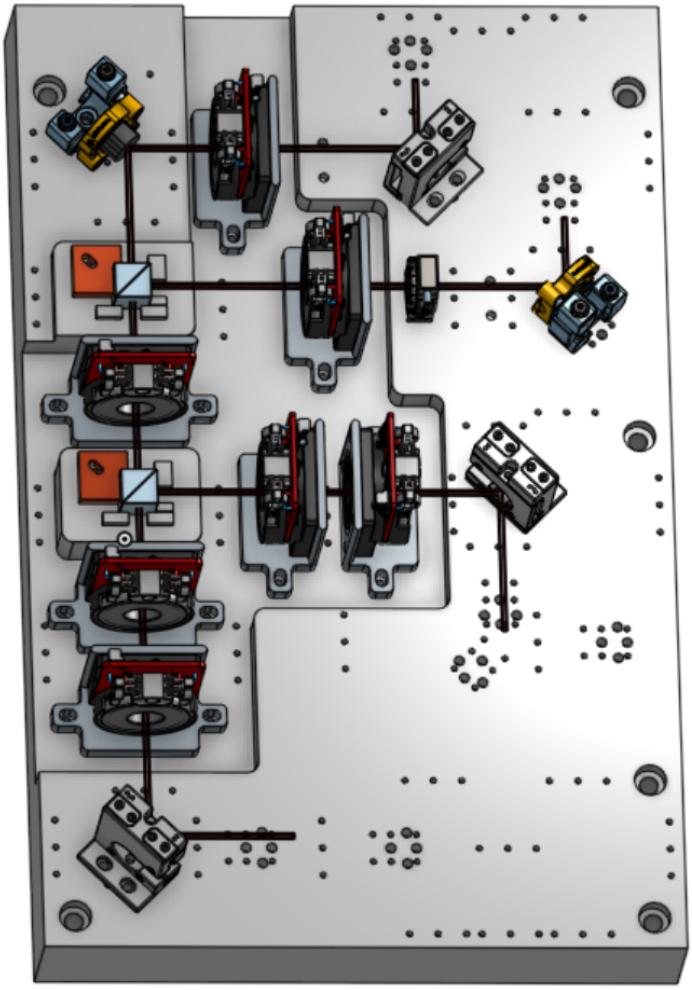


Figure 22: Example Assembly

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